Hugo Jeannin

PhD Student · Applied Mathematics

Université de Bordeaux, France

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I am a third-year PhD student in applied mathematics at the University of Bordeaux. I am looking for a postdoctoral position starting in October 2027.

Research interests: Random matrix theory, high-dimensional statistics and machine learning.

Education

University of Bordeaux — *Bordeaux, France*

PhD Applied Mathematics — *2024 - present*

- Supervisors: Pascal Vallet, Guillaume Ginolhac and Florent Bouchard
- Subject: High-Dimensional Statistics and Random Matrix Theory: Applications to Hyperspectral Imaging
- Developed methods for signal subspace dimension estimation in high-dimensional Gaussian models
- Studied covariance-based classification using random matrix corrections

CentraleSupélec, University Paris-Saclay — *Gif-sur-Yvette, France*

MS Applied Mathematics — *2019 - 2023*

- Probability theory, statistics, optimisation, machine learning and PDE

Lycée Lakanal — *Sceaux, France*

Preparatory class — *2017 - 2019*

- Mathematics, theoretical physics and chemistry

Publications

Submitted manuscripts

Hugo Jeannin, Pascal Vallet, Florent Bouchard, Guillaume Ginolhac, Audrey Giremus. *A new approach for estimating the number of spikes in high-dimensional Gaussian models*. Submitted to IEEE Transactions on Signal Processing. DOI (arXiv): 10.48550/arXiv.2610.18059

Hugo Jeannin, Pascal Vallet, Guillaume Ginolhac, Florent Bouchard. *Pooled Covariance Whitening and Random Matrix-Corrected Grassmann Distance for Spiked Covariance Classification*. Manuscript under review.

Published

Philippe Formont, **Hugo Jeannin**, Pablo Piantanida, Ismail Ben Ayed. *A Strong Baseline for Molecular Few-Shot Learning* (*2025*). Transactions on Machine Learning Research. DOI (arXiv): 10.48550/arXiv.2404.02314

Hugo Jeannin, Florent Bouchard, Pascal Vallet, Guillaume Ginolhac, Audrey Giremus. *On the endmembers detection for hyperspectral imaging in the high-dimensional regime* (*2025*). EUSIPCO. DOI: 10.23919/eusipco63237.2025.11226606

Professional Experience

European Space Agency (ESA), Φ-lab — *Frascati, Italy*

Visiting Researcher — *November 2026 – Present*

- Training foundation models for hyperspectral imaging

Université Grenoble-Alpes — *Grenoble, France*

Research Engineer — *February 2024 – August 2024*

- Studied partial least squares (PLS) in high-dimensional settings, with applications to genomic data analysis

International Laboratory on Learning Systems (ILLS) — *Montreal, Canada*

Research Internship — *July 2023 – December 2023*

- Research on graph neural network methods for few-shot molecular classification

RTE (French power grid network) — *Paris, France*

Research Internship — *February 2022 – June 2022*

- Developed a new operations research method to optimize the replacement of components on the network

Technical Skills

Programming and tools: Python, R, MATLAB, Git.
Languages: French, English (C1), German (B2).